

# A COMPARATIVE STUDY OF CNN-BASED AND HYBRID CLASSIFICATION MODELS FOR FLOWER SPECIES RECOGNITION

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## Problem Statement

- High visual similarity among flower species makes accurate classification difficult.
- Different deep learning models vary in feature extraction and generalization performance.
- A systematic comparison is needed to identify the most effective model.



## Models Evaluated

1

### Baseline: Model 1 (From Scratch)

- Demonstrates necessity of transfer learning
- Insufficient capacity for 102 fine-grained classes

2

### Best Performer: Model 2 (ResNet50 + Fine-tuning)

- Highest validation accuracy: 95.84%
- Two-phase training: Feature extraction + Fine-tuning top 30 layers
- Strong regularization (L2, Dropout, BatchNorm, Early Stopping)

3

### Most Efficient: Model 3 (Hybrid Approach)

- No neural network training required (only SVM)
- Fast training after feature extraction
- 90.46% accuracy with minimal computational cost

4

### Most Efficient: Model 3 (Hybrid Approach)

- Smallest gap (1.2%) between training and validation
- Frozen backbone prevents overfitting
- Early stopping at epoch 45 ensures efficiency

## Results

- Model 1 (CNN Scratch): Val Acc  $\approx$  52% [underfits, limited capacity]
- Model 2 (ResNet50 + CNN): Val Acc  $\approx$  96.6% [highest accuracy, fine-tuned top 30 layers]
- Model 3 (MobileNetV2 + SVM): Val Acc  $\approx$  90.5% [PCA + linear SVM reduced overfitting]
- Model 4 (ResNet18 + MLP): Val Acc  $\approx$  88% [frozen backbone, well-balanced performance]

## Analysis

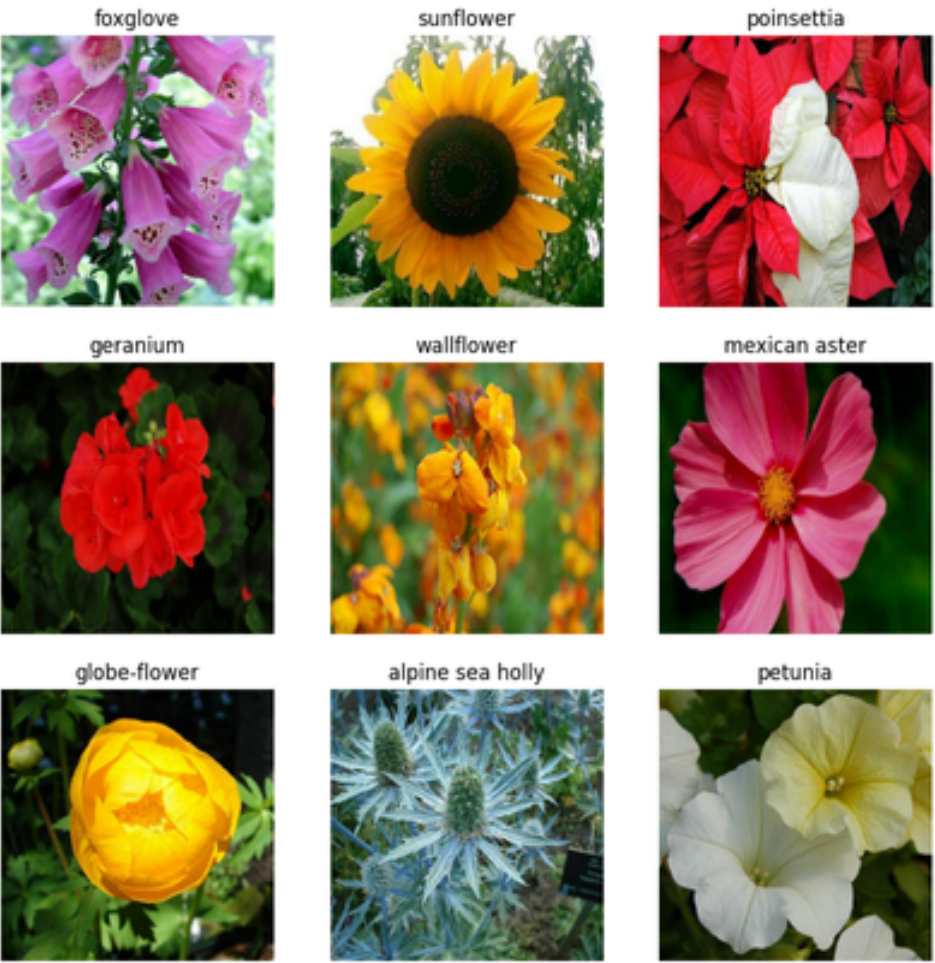
- Transfer learning consistently outperforms scratch CNN [pre-trained weights capture edges, textures, patterns]
- Fine-tuning top layers improves domain-specific feature recognition [Model 2]
- Controlling overfitting is key for both CNNs and classical classifiers [Model 3 PCA/SVM, Model 4 EarlyStopping]
- Lightweight models + frozen backbones offer fast, stable, and memory-efficient solutions [Models 3 & 4]

## Conclusion

- Pre-trained models essential for fine-grained classification with limited data [ImageNet weights boost performance]
- Fine-tuning top layers + regularization gives best results [Model 2]
- Hybrid approaches balance efficiency and accuracy [Model 3]
- EarlyStopping is reliable to prevent overfitting across all models
- Careful model selection + leveraging pre-trained knowledge is crucial for distinguishing visually similar flower species

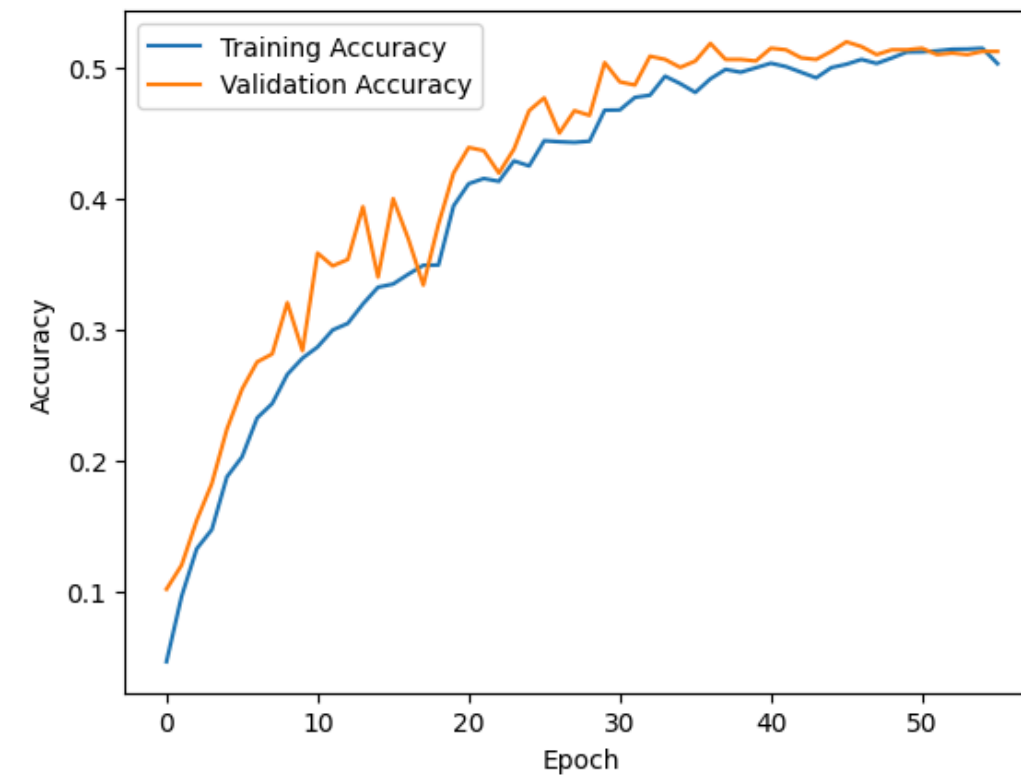
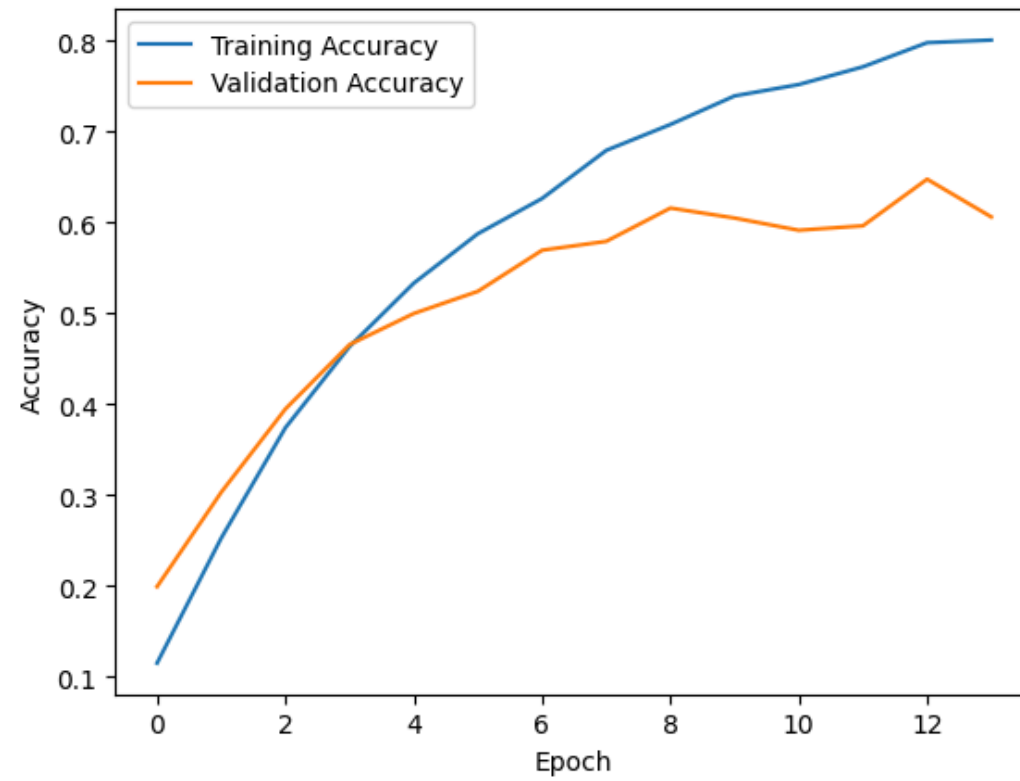
## Objective

- Develop and evaluate models for classifying 102 flower categories.
- Assess performance using accuracy metrics and visual prediction analysis.





## Scratch CNN

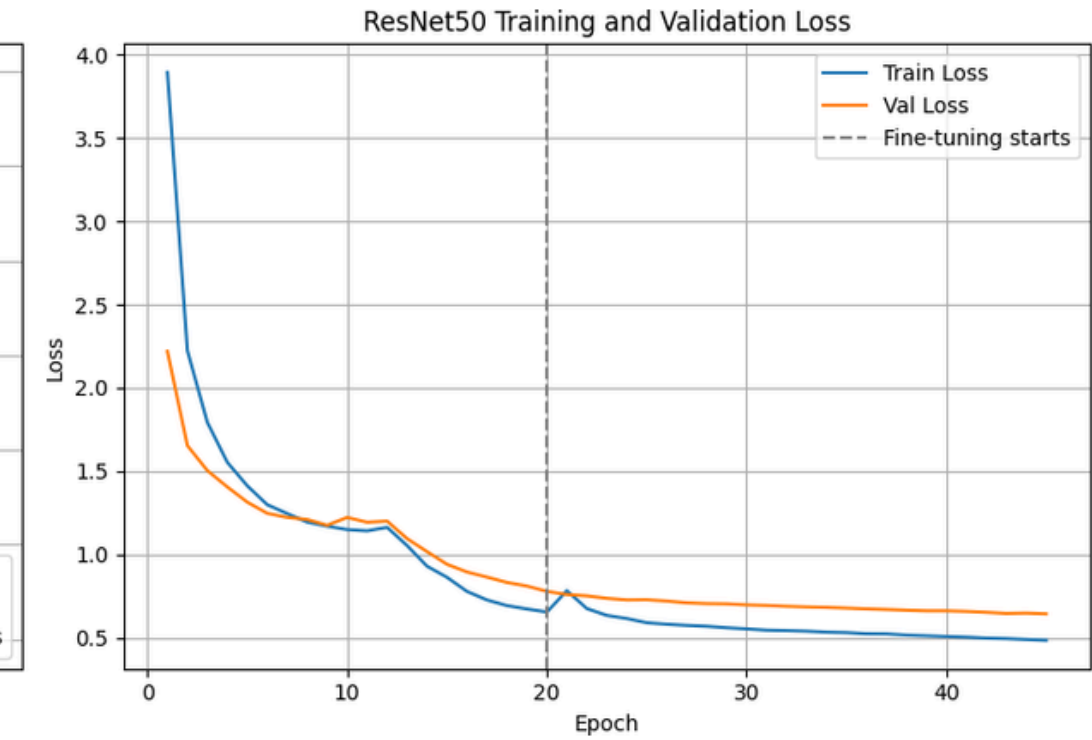
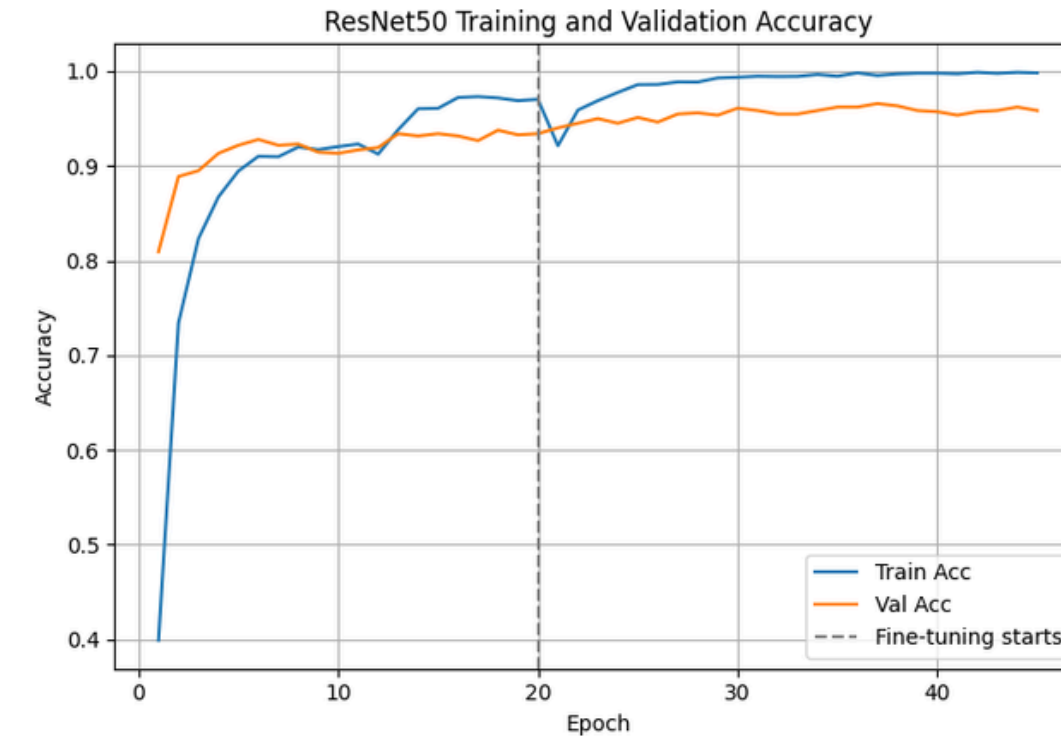


**Before** and **After** of the scratch CNN model. **What's the difference ?**

- L2 regularization
- Dropout layers
- Hyperparameter tuning
- Learning rate rescheduling

However, the accuracy is still low. **Whats the fix ?**

## Resnet50 + CNN

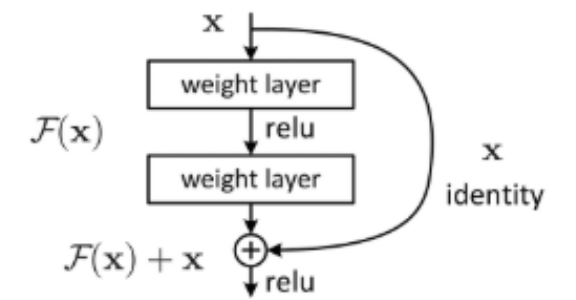


What is **ResNet50**?

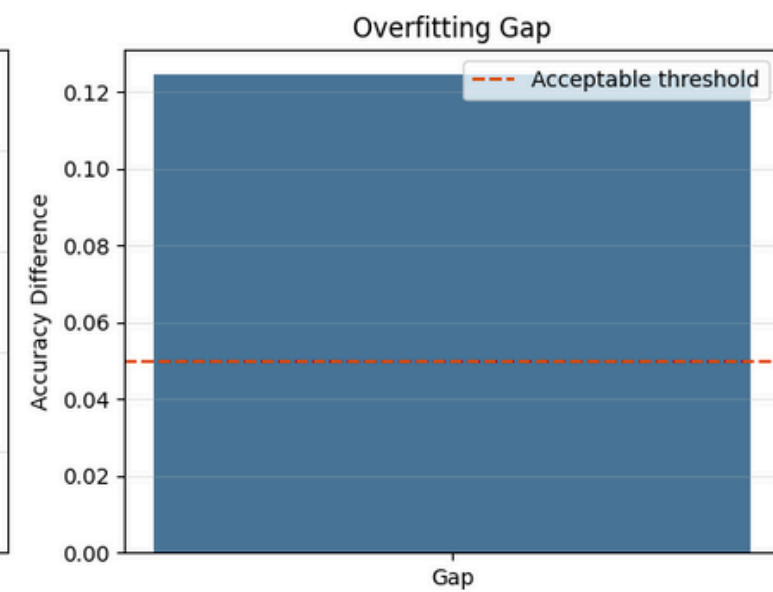
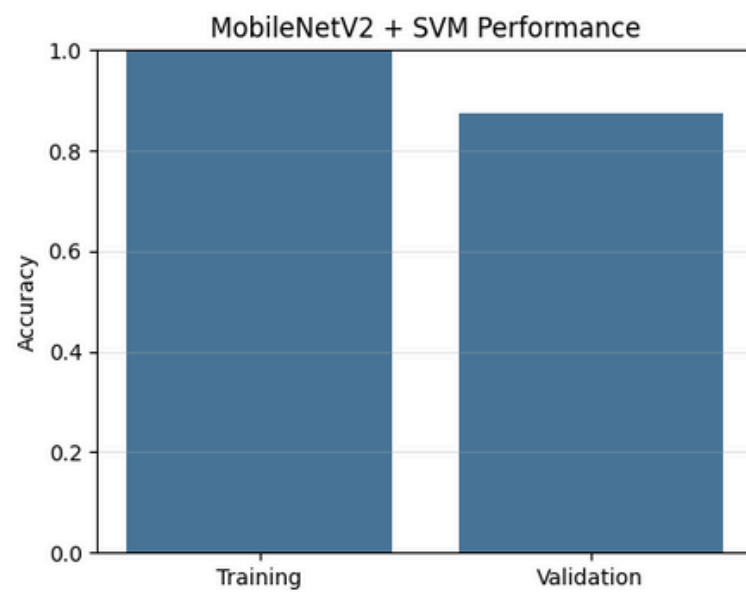
- ResNet50 is a pre-trained Classification CNN model with a 50 layer architecture.
- ResNet50 introduces skip connections that allow network to learn residual mapping instead of full mapping.
- Allows for better gradient flow and stable optimization.

Final validation accuracy is 96%, while training accuracy is 99%.  
Final result suffered from slight **overfitting**, but delivered a very high accuracy

### Residual Networks

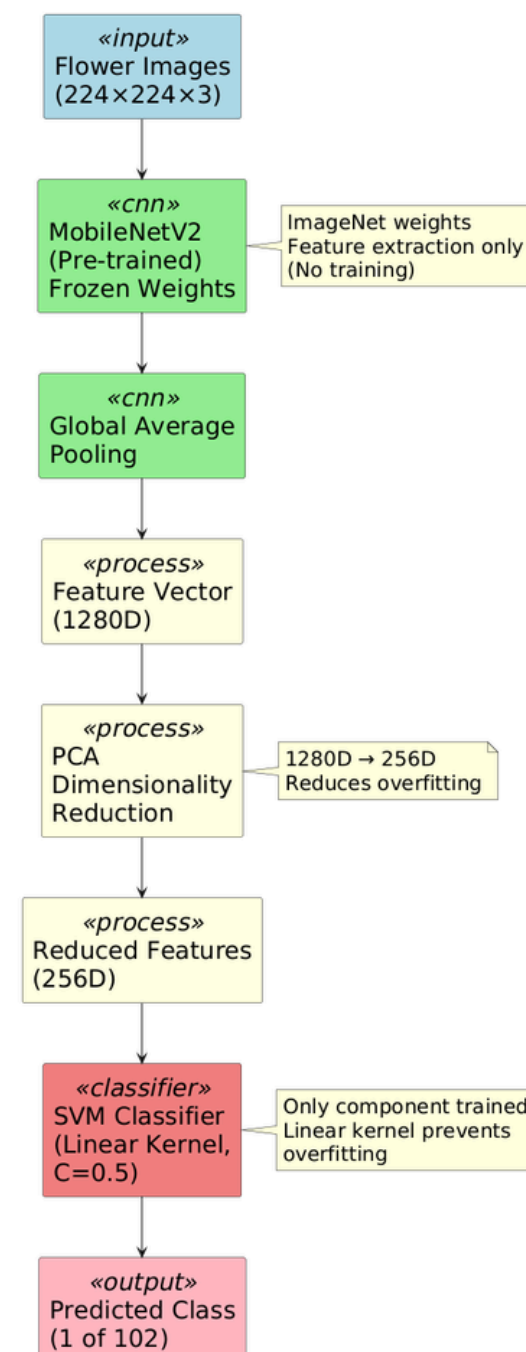


## MobileNetV2 + SVM

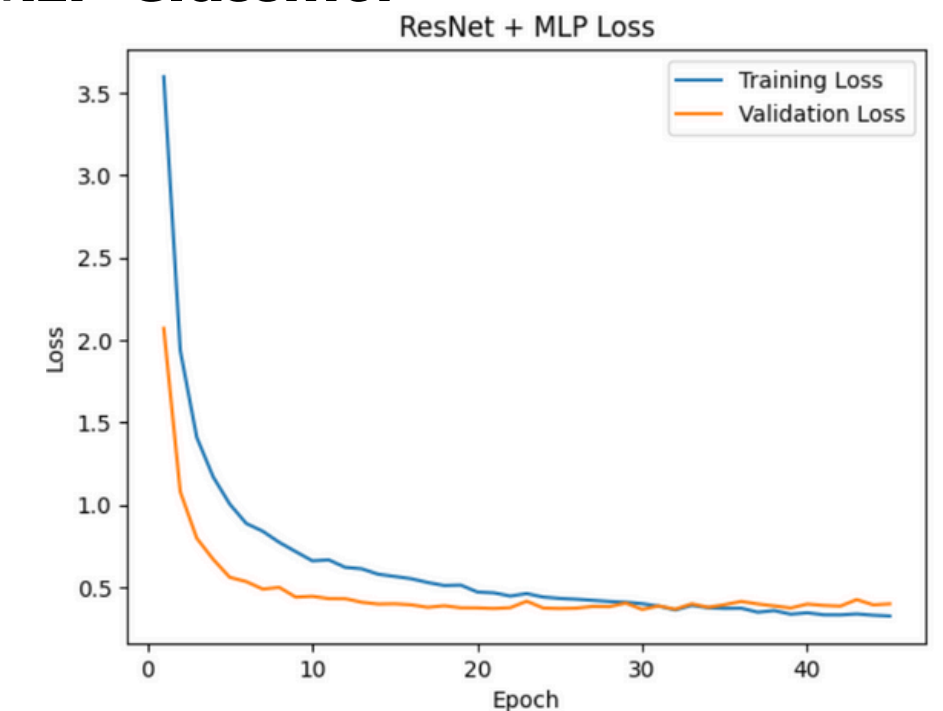
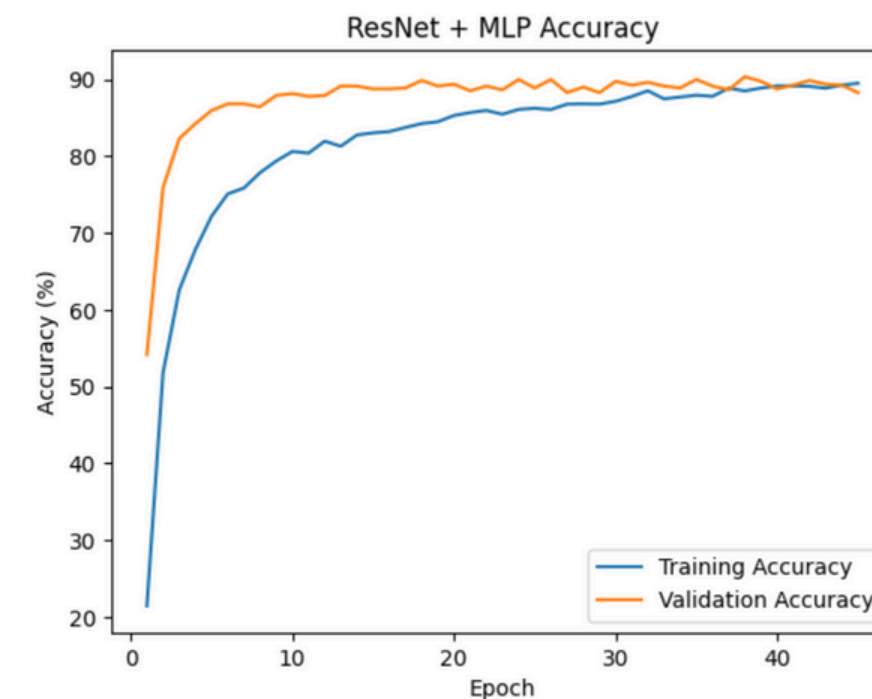


- **Approach:**
  - Pre-trained MobileNetV2 as frozen feature extractor (ImageNet weights)
  - Extract 1280-dimensional feature vectors via global average pooling
  - SVM classifier with RBF kernel for multi-class classification
- **Solution to reduce overfitting:**
  - Dimensionality Reduction: PCA (1280D → 256D)
  - Simpler Kernel: Linear SVM (instead of RBF)
  - Stronger Regularization: C = 0.5 (reduced from 10)

**Slightly overfitting** even after PCA + Regularization (Train: 100%, Val: 90.46%, gap 9.54%)



## ResNet18 + MLP Classifier



What is ResNet18 and MLP?

- **ResNet18:**
  - 18-layer deep Convolutional Neural Network
  - Pre-trained on ImageNet dataset
  - Uses residual connections to extract visual features
  - Frozen = Not retrained, preserves learned knowledge
- **MLP (Multi-Layer Perceptron):**
  - Simple fully-connected neural network
  - 2 layers: Hidden layer (256 neurons) + Output layer (102 neurons)
  - Acts as classifier for the 102 flower species
  - Trainable = Only this part learns from flower data

Well-balanced model with **no overfitting** (Train: 89.48%, Val: 88.26%, gap only 1.22%)

